

HTML & CSS

What is HTML?



- It's the **markup language** to create web pages.
- It stands for HyperText Markup Language.
- It uses "tags", which consists of angle brackets: <html>
- It provides the structure of the web page
- The latest version is 5: HTML 5

Basic HTML Web Page Structure

```
<!DOCTYPE html>

<html>

<head>
    <meta charset="utf-8">
    <title> Page title goes here </title>
</head>

<body>
</body>

</html>
```

Basic HTML Tags

HTML Tag	Description
<h1> to <h6>	Headings
	Bold text
<a href="" >	Links
<img src="">	Adds an image **(There is NO closing tag)**
	Unordered, Ordered lists
	List item
<p>	Paragraph
 	Line break **(NO closing tag)**
<hr>	Horizontal rule **(NO closing tag)**
	Creates a rectangular “division” or area

HTML Elements

HTML elements include:

a) the start tag,
b) some content, and
c) the end tag.

Example of an HTML Element:

<h1> Welcome! </h1>

HTML Elements can be styled with CSS!

What is CSS?



Cascading Style Sheets (CSS) is a language that defines the **layout** and **formatting** of a web page.

- **Layout:** Positioning, alignment, margin, padding, etc.
- **Formatting:** Color, size, font family, background, etc.

Ways to Apply Styles

1) Inline

- The style is included within a specific HTML element.
- Not a good practice, its use is discouraged because code gets very messy
  Example: <h1 style="color:red" />

2) Embedded/Internal

- Styles are included within the <head></head> section
- Styles are surrounded by the <style></style> tags
- Their scope is limited just to the current web page

3) External

- Best practice
- Styles are located in an external file and included using:
  <link href="path_to_styles.css" rel="stylesheet" type="text/css" />

CSS Rule Anatomy

CSS Rule: A rule consists of a selector and its declaration

```
<style>

    h1
      {
        color:        #0000FF;
        text-align: center;
      }

</style>
```

Selector (HTML element)

Declaration

Each **declaration** consists of two parts:
A property and a value.
They are separated by a colon.

CSS Selector: Tag name

HTML Elements can be selected based on **Tag name**, **Id**, and **Class**

Example of a selector using **Tag Name**

```
li  {
      color:   red;
 }
```

CSS rule targeting the “li” tag

```
<li> Monday </li>
<li> Tuesday </li>
<li> Wed </li>
```

HTML Code

Monday
Tuesday
Wed

Output

The styles are applied to
ALL elements with the tag

CSS Selector: ID Attribute

HTML Elements can be selected based on **Tag name**, **Id**, and **Class**

Example of a selector using **Id attribute**

```
#dayoff  {
      color:   red;
 }
```

CSS rule targeting an “id”

Observe that the id must be preceded with a hashtag #

```
<li> Monday </li>
<li> Tuesday </li>
<li id="dayoff"> Wed </li>
```

HTML Code

No hash tag

Id value **must be unique** on the entire page

Monday
Tuesday
Wed

Output

The style is applied to ONE element

CSS Selector: Class Name

HTML Elements can be selected based on **Tag name**, **Id**, and **Class**

Example of a selector using **a Class Name**

```
.dayoff  {
      color:   red;
 }
```

CSS rule targeting a “class”

Observe that the class must be preceded with a dot

```
<li class="dayoff">Monday </li>
<li> Tuesday </li>
<li class="dayoff">Wed</li>
```

HTML Code

No hash tag

Id value **must be unique** on the entire page

Monday
Tuesday
Wed

Output

The style is applied to those elements with the same class name